Retail Sales Performance Analysis Using SQL

1. Project Overview

This project analyzes retail sales transaction data using SQL to uncover business insights related to sales performance, customer behavior, product profitability, and regional trends.

The objective is to help business stakeholders make data-driven decisions by identifying key growth opportunities and operational challenges.

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2. Tools and Technologies

- MySQL
- SQL
- GitHub
- VS Code

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3. Dataset Description

The dataset contains retail sales transaction records including:

- Order Information
- Customer Information
- Product Information
- Sales Revenue
- Profit
- Quantity Sold
- Discounts
- Geographic Information

The dataset was imported into MySQL and analyzed using SQL queries.

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4. SQL Concepts Applied

Basic SQL

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING

Intermediate SQL

- Joins
- Subqueries

Advanced SQL

- Common Table Expressions (CTE)
- Window Functions
- Stored Procedures
- Stored Functions
- Ranking Functions

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5. Business Questions Solved

1. What is the total sales revenue?
2. What is the total profit generated?
3. Which product categories perform best?
4. Which products generate the highest sales and profit?
5. Who are the top customers?
6. Which regions contribute the most revenue?
7. How do discounts affect profitability?
8. Which products generate losses?
9. Who is the highest-value customer in each region?
10. What are the top products within each category?

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6. Key Findings

Sales Analysis

- The business generated significant revenue across multiple product categories.
- Average order value provided insight into customer spending behavior.

Product Analysis

- A small group of products contributed a large portion of sales revenue.
- Certain products generated negative profits despite strong sales.

Customer Analysis

- Top customers contributed a significant share of total revenue.

- Customer spending was concentrated among a limited number of high-value customers.

Regional Analysis

- Sales and profitability varied across regions.
- Certain regions consistently outperformed others.

Profitability Analysis

- Higher discounts negatively impacted profit margins.
- Loss-making products were identified for further review.

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7. Business Recommendations

1. Focus marketing efforts on top-performing categories.
2. Develop loyalty programs for high-value customers.
3. Reduce excessive discounting on low-margin products.
4. Increase inventory planning for high-demand products.
5. Investigate causes of losses in underperforming products and regions.
6. Use customer segmentation to improve targeted marketing.

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8. Conclusion

This project demonstrates the use of SQL for business analysis by transforming raw transactional data into actionable insights. Advanced SQL techniques including Joins, Subqueries, CTEs, Window Functions, Stored Procedures, and Stored Functions were used to support business decision-making and identify growth opportunities.

Author

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